
OPCP - Deployment Guide - Workloads

OVHcloud



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1 Disclaimer

TLP:AMBER+STRICT

restricts sharing to the organization only.

2 Revision history

Revision	Date	Author(s)	Modification
1	11/05/2026	Olivier BEDOUET-Solution Architect	Initial version
2	02/06/2026	Olivier BEDOUET-Solution Architect	- Fix networks name - add 2 workload domain design- add list of VMware public URLs - add VCF hosts connection detail
3	12/08/2026	Pierre-Yves AILLET-Tech Lead	- Update the documentation regarding the deployment without L3 routers

3 Introduction

3.1 Object of this document

This document serves as a guide for deploying workloads like VMware vSphere on OVHcloud OnPrem Cloud Platform within a customer environment. It describes supported design and customer tasks.

3.2 Perimeter

OVHcloud products can be delivered on a customer location, providing the capability to completely isolate and dedicate an OVHcloud environment On Premise.

3.3 Public

Document is to be used by Architect, for infrastructure design, and by System Engineer, for the technical implementation.

4 References

4.1 Applicables documents

Reference	Title
how-to-deploy-global	OPCP - Deployment Guide
how-to-deploy-compute	OPCP - Deployment Guide - Compute
how-to-deploy-network	OPCP - Deployment Guide - Network

5 VMware vSphere - Standalone esxi deployment

vSphere from VMware can be deployed on OPCP using the following recommendations.

Such deployment is only valid using VMware vSphere ESXi 7.0 and 8.0. For VCF or VWF 9, a VCF Operations instance and vCenter instance is needed.

5.1 Image creation

The ESXi installer provided by VMware is not compatible with Glance images support. So you need to create a supported image file of ESXi by installing inside a virtual machine running with QEMU or KVM.

The image can be created using the two following guides:

- <https://github.com/virt-lightning/esxi-cloud-images>
- <https://github.com/goneri/esxi-cloud-init>

5.2 Network Configuration

On OPCP, the recommendation is to configure the 4 network interfaces in two groups:

- A first group for a distributed vswitch named “dvs-Admin”, dedicated for VMkernel vNIC, using active-passive setup
- A second group for a distributed vswitch named “dvs-Compute”, dedicated for Virtual Machines, using a LACP portgroup

All the vlans tagged from these dvs will be encapsulated in Neutron networks using Q-in-Q tunneling, it does not need to set up Trunk ports on Openstack. Adding virtual network on a dvs is done simply by adding a port group.



Then proceed with components configuration:

- Needed VMkernel interfaces on dvs-Admin
- Vlans related to virtual machines on dvs-Compute

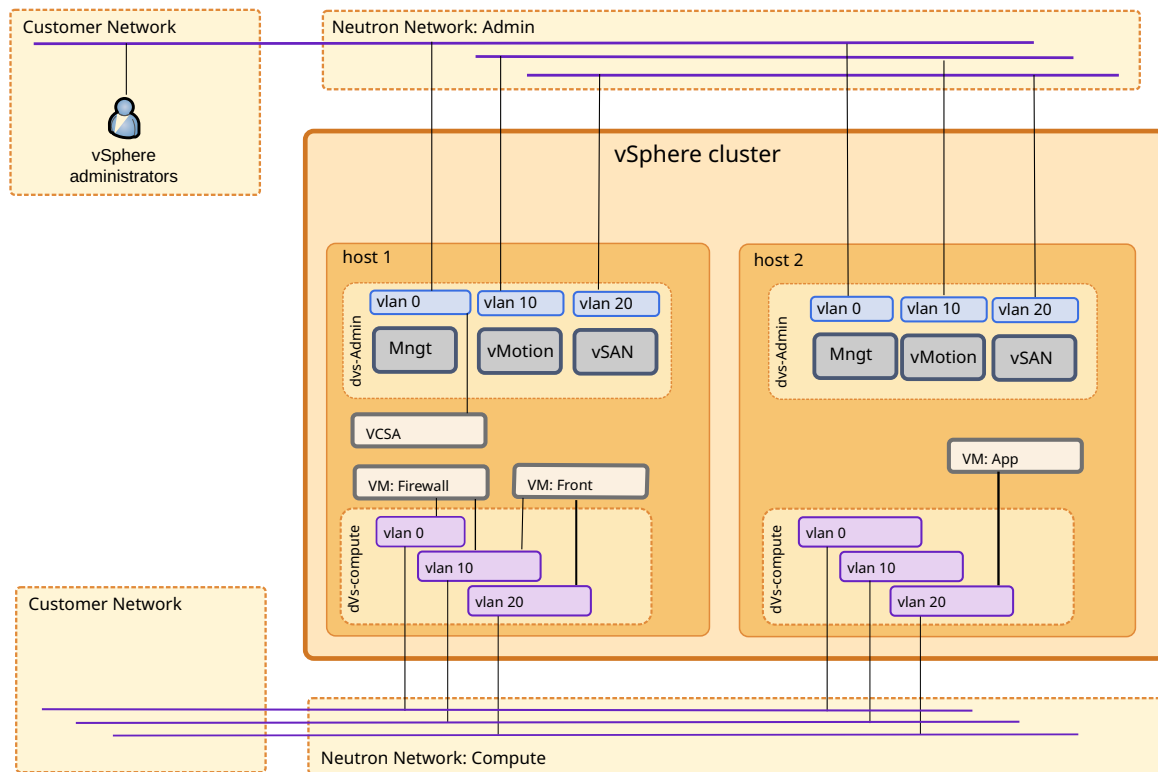


Figure 2: vSphere Distributed vSwitches Distribution

Both Neutron networks are completely isolated from each other, allowing the re-use of the same vlan ids between distributed vSwitches. It is advisable to connect these networks to different network infrastructure to avoid vlan overlap.

Note: Vlan 0 match untagged traffic on the network.

5.3 Storage Configuration

References servers hardware configuration often include two SSD for system installation. For operating systems like Linux, we recommend a software RAID1 configuration with the two SSD. As ESXi do not support software RAID, the recommended setup is:

- install ESXi on the first SSD
- format the second SSD as Local Datastore

The Local Datastore is useful for ESXi upgrades and also provide a storage space for swap files.

To run production workload, network storage like NFS or iSCSI with an external NAS appliance is possible using EDGE devices. Please refer to the document “how-to-deploy-network”.

vSAN is also another storage solution using the 25GbE interfaces on the dVs-admin.

6 VMware vSphere - Using vSphere Cloud Foundation by OVHcloud

6.1 Requirements

- The deployment is a VSAN Only deployment. VSAN ESA.
 - The Hardware has to be compliant with ESA.
- Internet access is required for VMware licensing purpose and VMware updates
- NTP
- DNS delegation is required to the DOMAIN provided to VCF

6.1.1 Global IP addresses requirements

Network	Prefix size
Management	/22
Management domain - VMotion	/24
Management domain - vSAN	/24
Management domain - Overlay	/24
vSphere admin network	/30 (minimum)

6.1.2 IP addresses requirements per Workload domain

Network	Prefix size
VMotion	/24
vSAN	/24
Overlay	/24

6.1.3 Public URLs required for Online Functionalities

Purpose	Source	Destination URL	Port
Binaries download for components	- VCF Installer - vCenter - VCF Operations - VCF Download Tool - VCF services runtime that hosts software depot	- dl.broadcom.com - eapi.broadcom.com - vvs.broadcom.com - vsanhealth.vmware.com	443
Binaries download for Supervisor services and VCF services	VCF services runtime that hosts software depot	projects.packages.broadcom.com	443
Telemetry data for the customer experience improvement program (CEIP)	* SDDC Manager * All VCF services runtime instances	vcasa.vmware.com	443
Compatibility data	- VCF Installer - SDDC Manager - VCF Download Tool	vvs.broadcom.com	443
vSAN HCL Data	- VCF Installer - SDDC Manager - vCenter - VCF Download Tool	- vsanhealth.vmware.com - eapi.broadcom.com	443
Licensing	VCF Operations	- vcf.broadcom.com - eapi.broadcom.com	443
Update Manager Download Service (UMDS)	- SDDC manager - VCF Download Tool	auth.esp.vmware.com	443

6.2 Limitations

Max hosts per domain	64
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Max hosts (global)	500
Max workloads domains	23
Max cluster per domain	1

6.3 VCF deployment principle

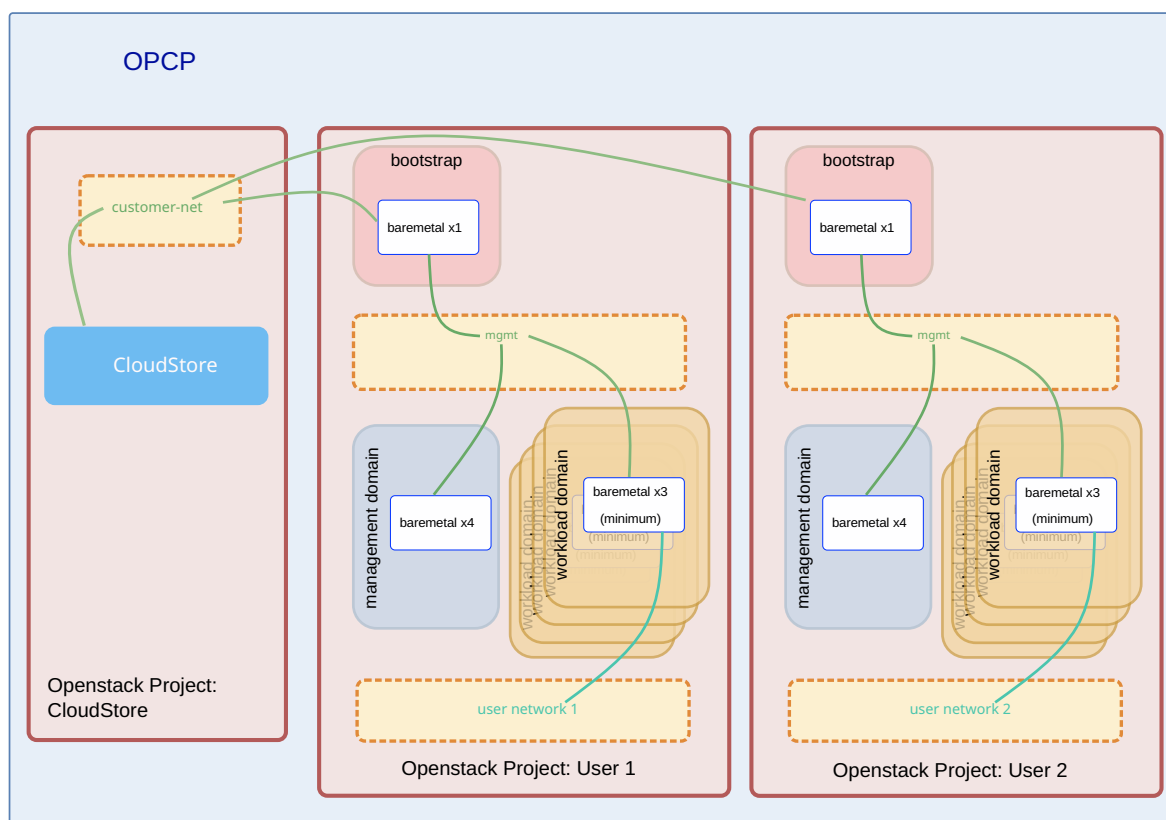


Figure 3: VCF deployment

Each VCF instance will be deployed inside a given Openstack project. Each VCF instance:

- Have a bootstrap server
- Need a minimum of 4x Baremetal for the Management Domain
- Can manage up to 23 workloads domain with a minimum of 3x Baremetal each

6.3.1 Bootstrap server

This is a dedicated Baremetal instance deployed from CloudStore. The purpose of this instance is to host the VCF Deployer and the OVH VCF Appliance in charge of automated operations. It is connected first with the CloudStore and after installation it is connected to the “mgmt” network.

6.3.2 Management Domain

Dedicated workload domain hosting:

- management components for all workload domains
 - vCenter
 - NSX Manager
 - SDDC Manager
- VCF fleet management components and instance level components

All components are connected to the management network.

It need a minimum of 4 hosts with the following requirements :

Host requirement	Value
CPU Cores	128
RAM Count (GB)	1024
Storage for vSAN (GB)	5855

The Management domain is only connected to the management network and its own VMotion, VSAN and Overlay networks.

6.3.3 Workload Domain

Host customer workloads. For the first one, a vCenter is deployed in the management domain. Each new workload domain will have a new vCenter deployment.

A NSX Manager is also deployed for each workload domain.

Each Workload Domain is connected to the management network and also to a customer network. This customer network need to be created before the deployment.

6.4 VCF deployment design

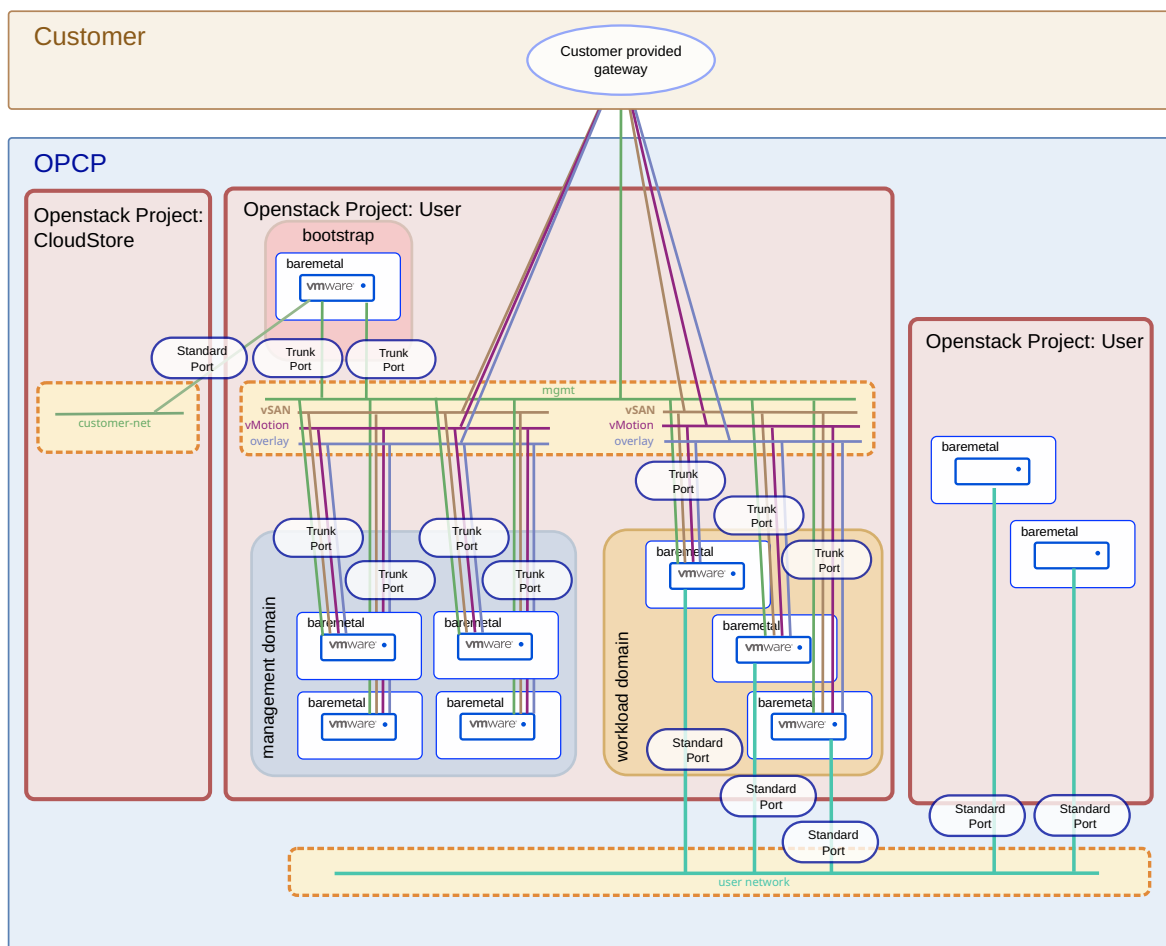


Figure 4: VCF High Level Design

This design presents a deployed VCF environment with a management domain and a workload domain. The workload domain is connected to "user network" which is also in use with another Openstack project.

Each domain has dedicated vSAN, vMotion and overlay networks and are connected to the mgmt network.

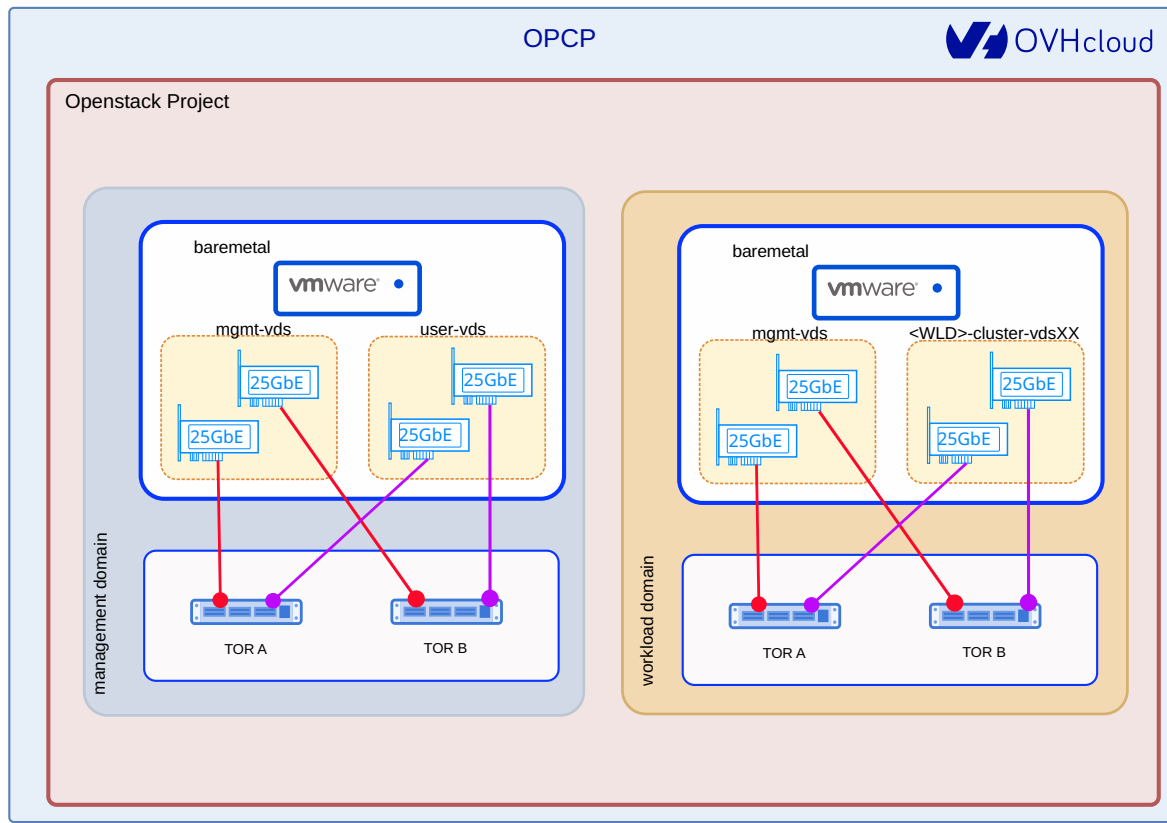


Figure 5: Detail of Host Network Configuration

6.4.1 Management network and VMkernel networks

Two VMnic are configured with a dedicated distributed vswitch named “mgmt-vds” using an Ironic portgroup in active-passive setup for the support of:

- Management network
- VMotion network
- VSAN network
- Overlay network

Each network use a dedicated Vlan tag on the interface which is configured on the TOR by using Openstack Neutron Trunk feature.

The configuration of the distributed vswitch and the correspond Trunk is not editable after the deployment.

6.4.2 Customer network

Two VMnic are configured with a dedicated distributed vswitch named “user-vds” using an Ironic portgroup in active-passive setup on the management domain and “WLD-cluster-vdsXX” with WLD the workload domain name and XX the workload domain id. The TOR is configured using a Q-in-Q tunnel so all Vlan configured on the dVs will be encapsulated.

The configuration of this distributed vswitch is up to the customer, OVHcloud deployment method will not edit any portgroup after installation.

6.4.3 Isolation between workloads domain

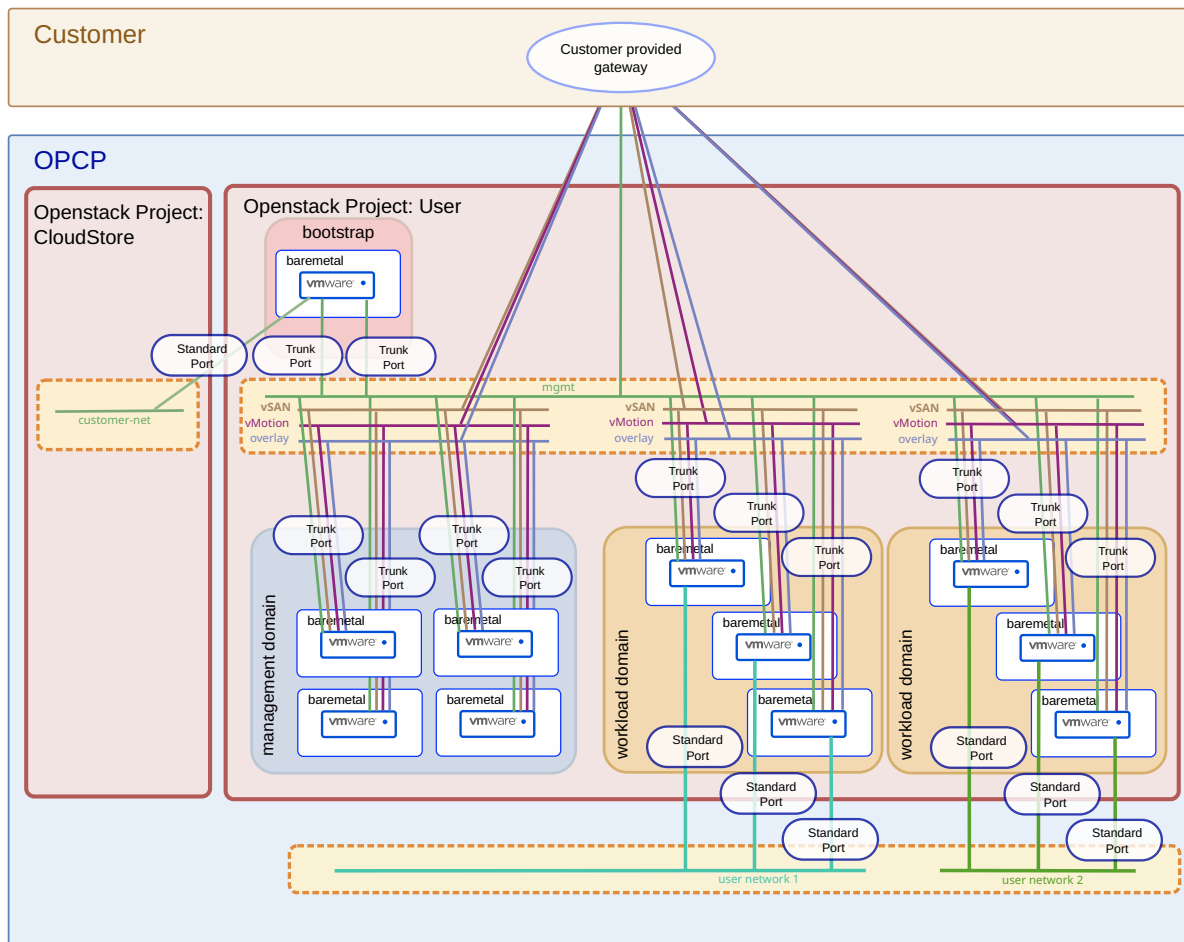


Figure 6: VCF Design with 2 workloads domain

The “mgmt” network is always shared between components of the same VCF deployment. The L3 router is also shared between vlans are dedicated per workload domain.

Customer network side can be dedicated per workload domain or shared.

6.5 VCF interfaces access

The management domain provide 3 interfaces:

- VCF Operations: <https://vcfops.DOMAIN>
- NSX Manager: <https://nsx.DOMAIN>
- vCenter Appliance (VCSA): <https://vcsa.DOMAIN>

For each workload domain:

- vCenter Appliance (VCSA): <https://vcsa-wldXX.DOMAIN>
- NSX Manager: <https://nsx-wldXX.DOMAIN>

With XX the number of the deployed domain (two digits format) like 01, 02, 10... up to 23.

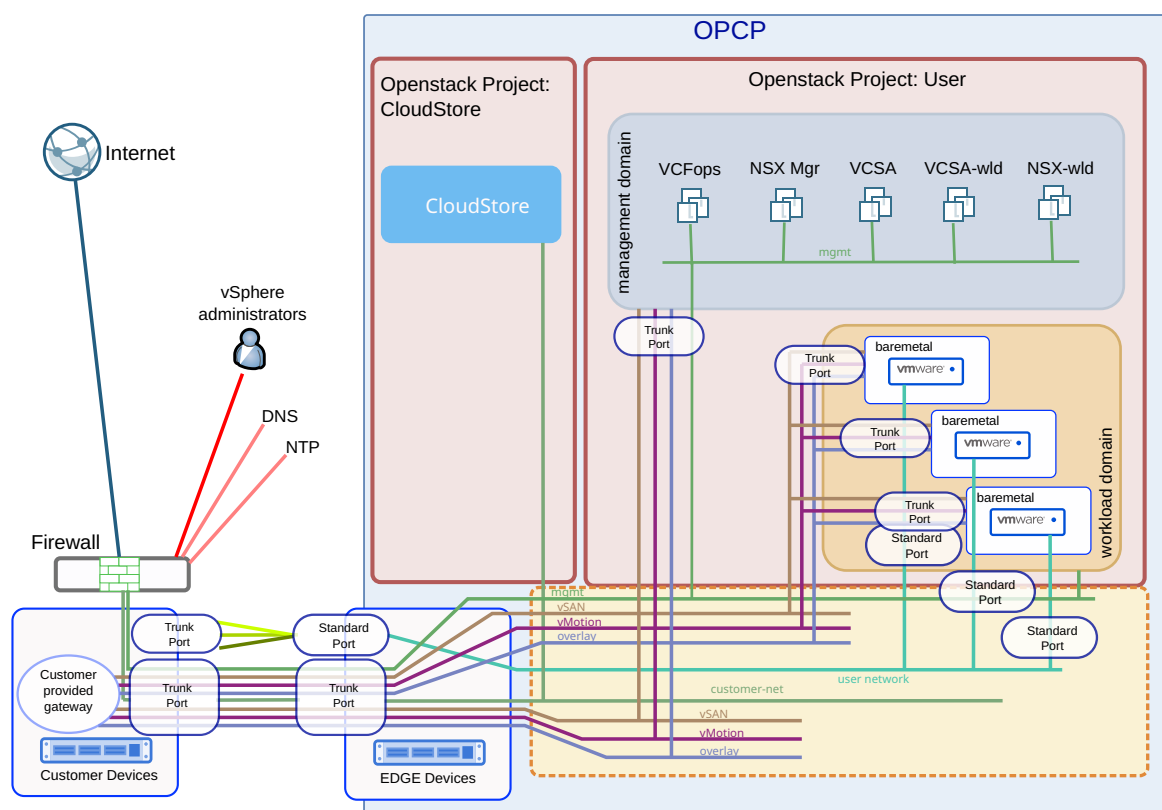


Figure 7: VCF Network Connectivity

Interfaces are reachable through the mgmt network through the default in-band of the EDGE devices. All VCF networks need to be extended to customer infrastructure:

- to provide a gateway in each configured subnet, mandatory for ESXi
- to provide connectivity with Internet through an external gateway for VCF services (updates and licences)
- to provide access to VCF interfaces
- to make a layer-2 network isolation between customer's infrastructure and VCF internal networks

Due to different encapsulation method, the user networks need to be connected through different interfaces on EDGE devices than the VCF "shared network".

All networks above need to be created and configured before the deployment.